MIL-D-13969(Ord)

21 February 1955

SUPERSEDING
Army 49-44
4 October 1945

MILITARY SPECIFICATION

DETONATOR, CONCUSSION TYPE, M1

1. SCOPE

1.1 This specification covers the M1 Concussion Type Detonator for use in detonating an explosive charge either under water or in air.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, drawings, and publications, and all drawings and specifications referenced thereon, of the issue in effect on date of invitation for bids form a part of this specification:

SPECIFICATIONS

MILITARY

MIL-G-2550 -General Specification For Ammunition Except Small Arms Ammunition

DRAWINGS

ORDNANCE CORPS

P-82109 -Detonator, Concussion Type M1 and M1 Inert, Assembly

PUBLICATIONS

ORDNANCE CORPS

ORD-SIP-M16-118 -Detonator, Concussion Type, M1. Loaded

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.-Material and parts shall be in accordance with applicable drawings and specification.

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3.2 The concussion detonators shall comply with all requirements specified on Drawing P-82109 and all drawings referenced thereon, and with all requirements specified in applicable specifications.

3.3 Internal air pressure.-The seal at the pellet sleeve and the seal at the concussion diaphragm of the detonator shall withstand without leakage an internal air pressure of 5 psi, minimum, for not less than 15 seconds when tested as specified in 4.8.

3.4 Firing pin.-The firing pin shall be neither thrown out of, nor be free to fall out of, the barrel when tested as specified in 4.9.

3.5 Sensitivity.

3.5.1 Functioning.-The concussion detonator shall fire the percussion cap when tested as specified in 4.10.1.

3.5.2 Safety.-The concussion detonator shall not fire the percussion cap when tested as specified in 4.10.2.

3.6 Workmanship.-All parts shall be free of chips, dirt, grease, rust and other foreign material. The cleaning method used shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents used.

4. QUALITY ASSURANCE PROVISIONS

4.1 Lot.-A lot shall consist of all concussion detonators assembled by one manufacturer under one contract in one unchanged process, in accordance with the same drawing, same drawing revision, the same specification, and same specification revision. Each lot shall contain:

a. metal parts or metal parts assemblies of one interfix lot number from one manufacturer.
b. primers from not more than two lots
c. blasting caps of one manufacturer bearing the same interfix number.

4.2 Sampling.-Number of samples, acceptance and rejection criteria used for determination of the lot acceptance for tests as specified herein shall be in accordance with Standard Inspection Procedure ORD-SIP-M16-118.

4.3 Inspection.-Inspection shall be as specified in Specification MIL-G-2550.

4.4 Firing pin pull test.-The firing pin of sample detonators shall be subjected to a force, by any satisfactory method, to determine compliance with the pull requirement specified on the applicable drawing.

4.5 Safety pin pull test.-The safety pin of sample detonators shall be subjected to a force, by any satisfactory method, to determine compliance with the resistance withdrawal requirement specified on the applicable drawing.

4.6 Delay pellet dissolve test.

4.6.1 Blue delay pellet.-Sample blue pellets shall be completely immersed in water and observation made to determine if the pellets comply with the dissolving time requirement specified on the drawing.

4.6.2 Yellow delay pellet.-Sample yellow delay pellets shall be completely immersed in water and observation made to determine if the pellets comply with the dissolving time requirement specified on the drawing.

4.7 Concussion diaphragm test.-Using any satisfactory method, sample concussion diaphragms shall be subjected to a weight-load capable of snapping the diaphragms to the reverse position. An additional weight-loads shall then be applied which is capable of snapping the concussion diaphragms back to their

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original free position. Following the application of each of the above weight-loads, observation shall be made to determine compliance with the concussion diaphragm requirement specified on the applicable drawing.

4.8 Internal air pressure test.-Each detonator shall be fitted to an air line by a special connection at the base and shall be subjected to an internal air pressure of 5 psi, minimum, while immersed in water or in alcohol. Observation shall be made to determine compliance with 3.3. Water or alcohol shall not be permitted to enter the device and all water and alcohol shall be dried from the outside before screwing the dummy base in the barrel.

4.9 Firing-pin test.-After assembly of each detonator, but just before the insertion of the dummy base, the safety pin shall be removed. Observation shall be made to determine compliance with 3.4. If the firing pin is thrown out of the barrel or if it is free to fall out of the barrel, the device shall be disassembled and all parts carefully checked before being tested again. If the firing pin is neither thrown out nor free to fall out, the device shall be disarmed and the safety pin reinserted.

4.10 Sensitivity test.

4.10.1 Functioning Test.-Sample detonators shall be tested, one at a time, by filling a wooden tank with inside dimensions of 48 inches by 24 inches by 18 inches with water to a depth of 12 inches. A special one-gram detonator, supplied by the Government, shall be lowered to a depth of 7 inches in the water. The concussion detonator with a firing-device base and percussion cap, but without a delay pellet or safety pin, shall also be lowered to a depth of 7 inches in the water and suspended at the level by a metal strap. The minimum distance between the special detonator and the diaphragm end of the concussion detonator shall be 5 inches. The special detonator shall be fired and observation made to determine compliance with 3.5.1.

4.10.2 Safety test.-Sample concussion detonators and special detonators shall be placed in the water tank as specified in 4.10.1, except that the detonators shall be between 1 and 1 1/2 inches apart and the safety pin on the concussion detonator shall be left in place. The special detonator shall be fired and observation made to determine compliance with 3.5.2.

5. PREPARATION FOR DELIVERY

5.1 Packing, labeling and marking.-Packing, labeling and marking shall be as specified on the applicable drawing.

5.2 Data cards.-Data card information shall be as specified in Specification MIL-G-2550.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the title, number, and date of this specification.

Notice.- This specification, together with specifications and drawings pertaining to it and bearing a "Notice" of similar restrictions, is intended for use only in connection with procurement by the United States Government, and shall not be reproduced either wholly or in part except when authorized in connection with Government procurement, nor be used for any other purpose except when specifically authorized by the Chief of Ordnance.

Custodian:

Army-Ordnance Corps

Other interest:

Army-E

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